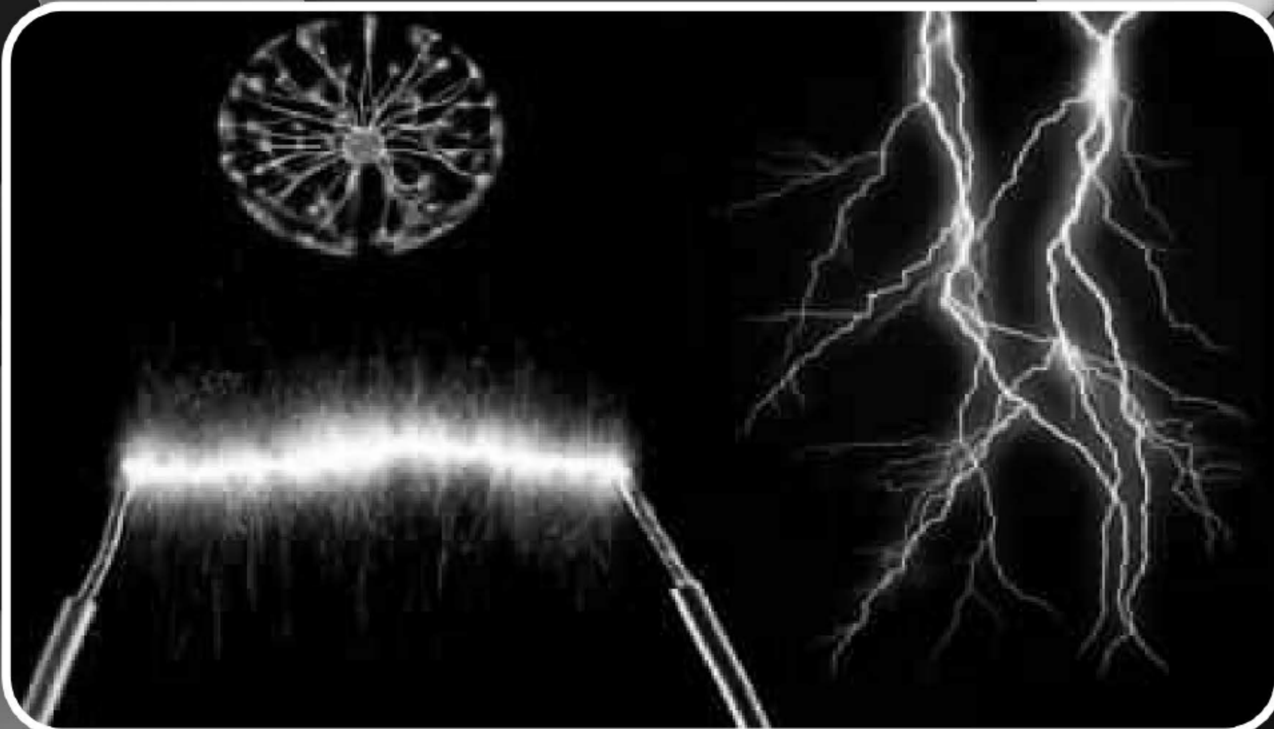
Unit 8

Electric Fields

Teaching Periods | 16 | Weightage % | 11



In this unit student should be able to:

- Define Electrostatic force
- Explain Coulomb's law
- Define the Coulomb's force in different mediums
- Solve problems using Coulomb's law
- Describe the concept of an electric field as an example of a field of force.
- Derive the expression $\vec{E} = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}$ for the magnitude of electric field at a distance 'r' from a point charge 'q'.
- Define electric field strength as a force part unit positive charge.
- Solve problems and analyze information using $\vec{E} = \frac{F}{q}$.
- Solve problems involving the use of the expression, $\vec{E} = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}$
- Describe the concept of electric dipole
- Calculate the magnitude and direction of the electric field at a point due to two charges with the same or opposite signs.
- Sketch the electric field lines for two point-charges of equal magnitude with the same or opposite signs.
- Describe electric flux.
- Explain electric flux through a surface enclosing a charge.
- Define absolute electric potential.
- Define potential difference and its unit.
- Solve problems by using the expression V=W/q.
- Calculate the potential in the field of a point charge using the equation,
$$V = \frac{1}{4\pi\varepsilon_0}\frac{q}{r}$$
- Show that the electric field at a point is given by the negative potential gradient at that point.
- Solve problems by using the expression, E=-V/d
- Define electron volt.

8.1.1 Electrostatic force:

In the early days, the early Greek philosophers has studied the physics of electrostatic by performing an activity; when amber is rubbed with silk or wool and brought near dust particles, the dust particles will jump to stick with the amber. This attraction between dust particles and amber is known as an electrostatic force.

The electrostatic force between two charges is analogy to the gravitational force. Two charged objects can (i) attract or (ii) repel each other with a force is known as electrostatic force.

Whereas the gravitational force is always attractive in nature. Furthermore, the electrostatic forces are short range forces as compare to the gravitational forces, but the electrostatic forces are strong forces than the gravitational forces.

The object having same nature of charges repel each other and having opposite nature attract each other with an electrostatic force.

DO YOU KNOW?

- Name: Charles-Augustin de Coulomb
- Birth Year: 1736
- Birth date: June 14, 1736
- Birth City: Angoulême
- Birth Country: France
- Gender: Male
- Best Known For: French engineer and physicist Charles de Coulomb made pioneering discoveries in electricity and magnetism, and came up with the theory called Coulomb's Law.

8.1.2 Coulomb's Law:

Charles-Augustin de Coulomb performed an experiment of torsion balance (see figure 8.1) in 1785 to measure the magnitudes of the electric force between charged objects. Once the spheres are charged by rubbing, the electric force between them is very large compared with the gravitational attraction, and so the gravitational force can be neglected. From Coulomb's experiments, we can generalize the properties of the electric force (sometimes called the electrostatic force) between two stationary charged particles.

- If two charged particles are brought near each other, they exert an electrostatic force on each other. The direction of the force vectors depends on the signs of the charges.

- If the particles have the same sign of charge, they repel each other. That means that the force vector on each is directly away from the other particle. Beside this,

Fig: 8.1
Coulomb's torsion balance.

- If the particles have opposite signs of charge, they attract each other and the force vector on each is directly towards the other particle as shown in figure 8.2.

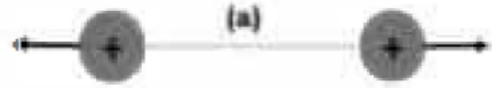


Fig: 8.2 (a) both are positive

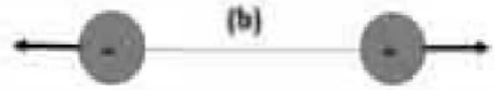


Fig: 8.2 (b) both are negative..

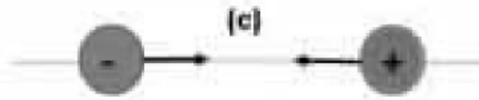


Fig: 8.2 (c) both are opposite

The illustration for two charges (q_1, q_2) and separation (r) between the charges is shown in figure 8.3.

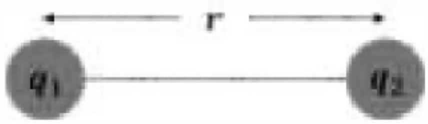


Fig: 8.3
Two charges q_1 and q_2 placed at a distance r from each other.

The electrostatic force F is directly proportional to the product of magnitudes of the charges q_1 and q_2. And inversely proportional to the square of the distance r between the charges and

$$\vec{F} \propto q_1 q_2$$

$$\vec{F} \propto \frac{1}{r^2}$$

This result is known as the inverse square law.

$$\cancel{\vec{F} = k}\frac{q_1 q_2}{r^2}$$

Where k is known as the Coulomb's constant and $k = \frac{1}{4\pi\varepsilon_0}$.

$$\vec{F} = \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{r^2} \qquad (8.1)$$

8.1.3 Coulomb's force in different medium:

The permittivity of air at standard pressure is only about 1.005 times that of ε_0. Therefore, it can be assumed in many cases that the values of ε_0 are equal for both vacuum and air. The permittivity of water is about eighty times that of a vacuum. Thus, the force between charges situated in water is eighty times less than if they were situated the same distance apart in a vacuum.

In general, the equation (8.1) can rewrite as

$$\vec{F} = \frac{1}{4\pi\varepsilon}\frac{q_1 q_2}{r^2}$$

where $\varepsilon = \varepsilon_r \varepsilon_0$ is the permittivity of the medium and ε_r is the relative permittivity.

DO YOU KNOW?

For your information

ε_0 is known as the permittivity of free space. if we suppose the charges are situated in a vacuum then the value of ε_0 is given as $\varepsilon_0 = 8.854 \times 10^{-12}\ C^2/N\,m^2$.
The value of Coulomb's constant in SI unit is given as
$k = 8.988 \times 10^9\ N\,m^2/C^2$

Material	Relative Permittivity
Vacuum	1.0000
Air	1.0006
PTFE, FEP (Teflon)	2.0
Polypropylene	2.20 to 2.28
Polystyrene	2.4 to 3.2
Wood (Oak)	3.3
Bakelite	3.5 to 6.0
Wood (Maple)	4.4
Glass	4.9 to 7.5
Wood (Birch)	5.2
Glass-Bonded Mica	6.3 to 9.3
Porcelain, Steatite	6.5

DO YOU KNOW?

Why do salt crystals dissolve in water?
A salt crystal is an ionic crystal. For example, in the case of sodium chloride (NaCl). The sodium ions (Na^+) and the chlorine ions (Cl^-) in a salt crystal are oppositely charged. The electrostatic forces between these ions hold the atoms together.. Note that Coulomb's law as stated above applies strictly to a vacuum. In terms of the force between two-point charges in air, the force is effectively the same as it would be in a vacuum.

Vector form of Coulomb's law:

Figure 8.4 illustrates the vector form of electrostatic force between two charges. The electric force F_{12} exerted by a charge q_1 on other charge q_2 is written as:

$$\vec{F}_{12} = \frac{1}{4\pi\varepsilon}\frac{q_1 q_2}{r^2}\hat{r}_{12} \qquad (8.2)$$

where $\hat{r}_{12}$ is a unit vector directed from q_1 toward q_2 as shown Figure 8.4. Because the electric force obeys Newton's third law,as equal and opposite direction; that is, $\vec{F}_{12} = -\vec{F}_{21}$

Fig: 8.4
The force F_{21} exerted by q_2 on q_1 is equal in magnitude and opposite in direction to the force F_{12} exerted by q_1 on q_2.

Worked Example 8.1

The electron and proton of a hydrogen atom (**figure A**) are separated by approximately $5.3 \times 10^{-11}\,m$. Find the magnitudes of (a) the electric force and (b) the gravitational force between the two particles. (c) What is your conclusion about these forces.

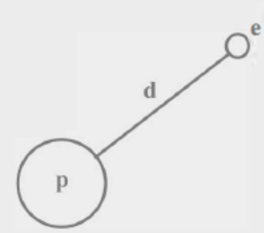


Figure A

Solution:

Step:1 Write down the known quantities and quantities to be found.

$r = 5.3 \times 10^{-11}\,m.$

(a) Charge on electron$= q_e = -1.602 \times 10^{-19}C,$
Charge on proton$= q_p = 1.602 \times 10^{-19}C,$
$k = 8.988 \times 10^9\ N\,m^2/C^2$
Electrostatic force$=F_e$=?

(b) Mass of electron $= m_e = 9.109 \times 10^{-31}\,kg$
Mass of proton $= m_p = 1.672 \times 10^{-27}\,kg.$
where $G = 6.67 \times 10^{-11}\ N\,m^2/kg^2,$
Gravitational force$=F_g$=?

Step:2 Write down the formula and rearrange if necessary.

(a) For electrostatic force $F_e = k\frac{q_e q_p}{r^2}$

(b) For gravitational force $F_g = G\frac{m_e m_p}{r^2}$

Step:3 Put the values in formula and calculate.

(a) $F_e = 8.988 \times 10^9 \ N\,m^2/C^2 \times \frac{-1.602\times 10^{-19}C \times 1.602\times 10^{-19}C}{(5.3 \times 10^{-11}\,m)^2}$

$$F_e = -8.2 \times 10^{-8}N$$

where negative sign indicates the force is attractive. The magnitude of this electrostatic force is $8.2 \times 10^{-8}N$.

(b) $F_g = 6.67 \times 10^{-11} \ N\,m^2/kg^2 \frac{9.109\times 10^{-31}\,kg \times 1.672\times 10^{-27}\,kg}{(5.3 \times 10^{-11}\,m)^2}$

$$F_g = 3.6 \times 10^{-47}N$$

Hence, the gravitational force of attraction between the particles is $3.6 \times 10^{-47}N$

(c) Conclusion:

The gravitational force between electron and proton is negligible as compared to the electric force, which implies that electric force is strong force.

Worked Example 8.2

Two positive charge of equal magnitude are placed of in vacuum at distance of 50cm and repel each other with the electric force of 0.1N. **(a)** Find the value of the charge. **(b)** Calculate the force between these two charges if they are places in an insulating liquid whose permittivity is five times that of a vacuum.

Solution:

Step:1 Write down the known quantities and quantities to be found.

$r = 0.5m$

$F = 0.1N$

$\varepsilon = 5\varepsilon_0$

(a) $q_1 = q_2 = q = ?$

(b) $F_{liquid} = ?$

Step:2 Write down the formula and rearrange if necessary.

(a) $F = k\frac{q_1 q_2}{r^2} \qquad \Rightarrow \quad q^2 = \frac{F\times r^2}{k}$

(b) $F_{liquid} = \frac{1}{4\pi\varepsilon}\frac{q_1 q_2}{r^2} = \frac{1}{5\times 4\pi\varepsilon_0}\frac{q^2}{r^2}$

$F_{liquid} = \frac{1}{5}F$

Step:3 Put the values in formula and calculate.

(a) $q^2 = \frac{0.1\times(0.5)^2}{8.988\times 10^9}$

$q = 1.7 \times 10^{-7}C = 0.17\mu C$

(b) $F_{liquid} = \frac{0.1N}{5} = 0.02N$

Hence, the magnitude of electric force for an insulating liquid is $0.02N$.